LLARS: An End-to-End Pipeline for Collaborative Prompting, Batch Generation, and Hybrid Evaluation

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Abstract

We demonstrate LLARS, an open-source platform that bridges the gap between domain experts and developers for systematic evaluation of LLM outputs. The platform provides three integrated modules that can be used together or independently: **Collaborative Prompt Engineering** lets domain experts and developers co-author prompts in a shared real-time workspace with version control and instant LLM testing. **Batch Generation** systematically produces outputs for all combinations of prompts, LLMs, and data items with built-in cost tracking. **Large-Scale Evaluation** supports human and LLM evaluators with live agreement metrics—applicable to outputs from any source. When combined, these modules form a complete pipeline from prompt development through rigorous assessment.

Demo: https://llars-project.github.io/demo

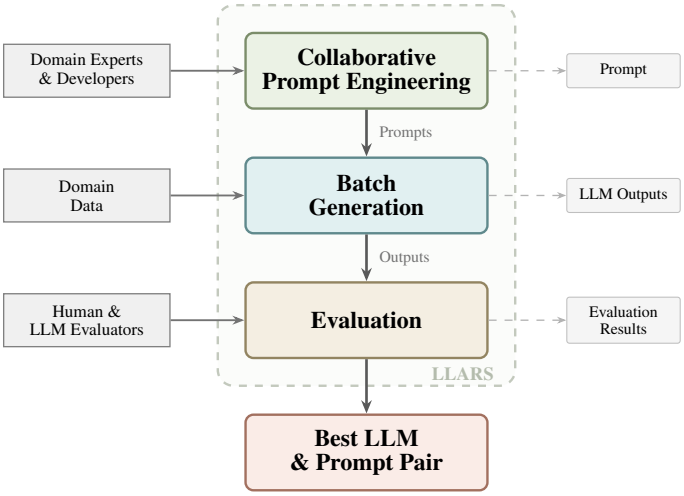

Figure 1: LLARS pipeline: domain experts and developers collaboratively develop prompts, generate outputs across LLMs, and evaluate results with human and LLM evaluators. Each stage supports export; the pipeline yields a validated model–prompt combination.

1 Introduction

Organizations increasingly build LLM-based systems for sensitive domains such as psychosocial counseling, legal analysis, and medical documentation. Developing these systems requires writing effective prompts, selecting suitable models, and rigorously evaluating outputs—tasks that demand both domain expertise and technical skill. Domain experts understand *what* constitutes quality in their field, while developers know *how* to structure prompts and configure LLM pipelines. Yet these two groups typically work in isolation: prompts are iterated in shared documents without version control, model outputs are evaluated ad hoc, and no single tool supports the full cycle from collaborative prompt development through systematic evaluation.

We present LLARS (LLM Assisted Research System), an open-source platform that bridges this gap with three integrated modules (Figure 1). *Collaborative Prompt Engineering* lets domain experts and developers co-author prompts in a real-time CRDT-based editor with version control and instant LLM testing—so both groups iterate in one workspace rather than exchanging documents. *Batch Generation* then systematically produces outputs for all combinations of prompts, models, and data items with cost tracking, enabling rigorous cross-model comparison. *Evaluation* lets human evaluators and LLM judges assess these outputs side by side—via rating, ranking, labeling, or pairwise comparison—while live inter-rater agreement metrics (Krippendorff's α, Cohen's κ, and others) reveal where human and machine judgment converge or diverge. Each module also operates independently with JSON/CSV export, and the platform connects to any OpenAI-compatible LLM provider including locally hosted models via Ollama.

Table 1 surveys twelve existing platforms along these dimensions. Agenta [Agenta AI, 2024] combines prompt management with LLM-as-judge scoring but does not offer real-time co-editing—essential when domain experts and developers jointly refine prompts—and lacks structured human evaluation campaigns with agreement metrics. The observability platforms Phoenix [Arize AI, 2024], Langfuse [Langfuse GmbH, 2024], and W&B Weave [Weights & Biases, 2024] provide dataset experiments and per-output feedback but focus on tracing and debugging rather than systematic batch generation or structured multi-evaluator campaigns. Chain-

Table 1: Comparison of GUI-first tools for prompt engineering, batch generation, and hybrid LLM output evaluation. ● = native support; ◐ = partial; ○ = not supported. **Collab.** = real-time collaborative prompt co-editing; **Batch** = systematic generation over prompts × models × data; **H-Eval** = structured human evaluation campaigns; **LLM-J** = LLM-as-judge; **IRR** = inter-rater reliability metrics; **E2E** = integrated end-to-end pipeline.

Tool	OSS	Collab.	Batch	H-Eval	LLM-J	IRR	E2E
LLARS	●	●	●	●	●	●	●
Agenta [Agenta AI, 2024]	●	◐	◐	◐	●	○	◐
ChainForge [Arawjo *et al.*, 2024]	●	○	●	○	◐	○	◐
Phoenix [Arize AI, 2024]	●	○	◐	◐	●	○	○
Langfuse [Langfuse GmbH, 2024]	●	○	◐	◐	●	○	○
W&B Weave [Weights & Biases, 2024]	●	○	◐	◐	●	○	○
Label Studio [HumanSignal, Inc., 2024]	●	○	○	●	◐	◐	○
Argilla [Argilla, Inc., 2024]	●	○	○	●	◐	○	○
LangSmith [LangChain, Inc., 2024]	○	○	◐	◐	●	○	◐
Braintrust [Braintrust Data, Inc., 2024]	○	○	◐	◐	●	○	◐
Maxim [Maxim AI, 2024]	○	○	◐	◐	●	○	◐
Vellum [Vellum AI, 2024]	○	○	◐	◐	●	○	◐

Forge [Arawjo *et al.*, 2024] supports visual multi-prompt comparison across models but does not include human evaluation and relies on custom metric functions instead of dedicated LLM judge prompts. The annotation platforms Label Studio [HumanSignal, Inc., 2024] and Argilla [Argilla, Inc., 2024] offer structured human assessment workflows—Label Studio also computes basic annotator agreement—but both require outputs to be generated externally, disconnecting evaluation from the prompt development process. The commercial platforms LangSmith [LangChain, Inc., 2024], Braintrust [Braintrust Data, Inc., 2024], Maxim [Maxim AI, 2024], and Vellum [Vellum AI, 2024] link prompt editing to evaluation with strong LLM-as-judge capabilities but are closed-source, lack collaborative editing, and do not include batch generation as an explicit pipeline stage.

By unifying collaborative prompt engineering, Cartesian-product batch generation, and hybrid human–LLM evaluation with comprehensive agreement analytics, LLARS enables a workflow where developers handle LLM integration, prompt versioning, and batch configuration while domain experts contribute quality criteria and evaluation judgments—both working in the same shared environment from prompt development through rigorous assessment.

2 Platform Overview

LLARS is a containerized web application built with Vue.js (frontend), Flask (backend), and MariaDB (storage). Real-time collaboration is provided by a dedicated Yjs [Jahns, 2023] server, and LLM access follows the OpenAI-compatible API structure, allowing teams to connect commercial providers (OpenAI, Anthropic, Google) as well as locally hosted models via Ollama. All interactions—prompt editing, generation progress, and evaluation updates—are synchronized via WebSocket.

2.1 Collaborative Prompt Engineering

Developing effective prompts for domain-specific LLM systems requires iterative refinement by those who understand the domain and those who understand prompt design. The prompt engineering module (Figure 2) addresses this through a real-time collaborative editor built on Yjs [Jahns, 2023], a CRDT framework that synchronizes edits at character granularity, merging concurrent changes without conflicts.

Users compose prompts as reorderable message blocks (e.g., system instructions, user templates) and define template variables using {{`variable`}} syntax. Variables are automatically extracted and displayed in a shared palette; during testing and batch generation, they are substituted with domain-specific data. Any collaborator can test the current prompt independently: clicking *Test* captures an atomic snapshot, fills variables with sample data, sends the prompt to a user-selected LLM, and streams the response—without interrupting concurrent editing. Multiple users can test against different models or data in parallel, enabling rapid exploration of prompt behavior across configurations.

A git-inspired version control system records each commit with multi-author attribution, enabling version comparison and rollback. Finalized prompts are exported as JSON or passed directly to batch generation.

2.2 Batch Generation

With prompts refined, the batch generation module scales output creation across combinations of data, prompts, and models (Figure 3). A guided wizard walks users through source selection, prompt and model configuration, and parameter tuning including budget limits. Sources can be existing evaluation scenarios, uploaded CSV or JSON files, or—for self-contained prompts that require no input data, such as creative text generation or instruction-following benchmarks—no source at all. The system generates the full Cartesian product: N data items $\times$ M prompts $\times$ K models. Template variables in prompts are automatically matched to data columns by name—a CSV column `content` fills every {{`content`}} placeholder across all selected prompts.

Users can register custom LLM providers with their own API keys; these models then appear throughout the

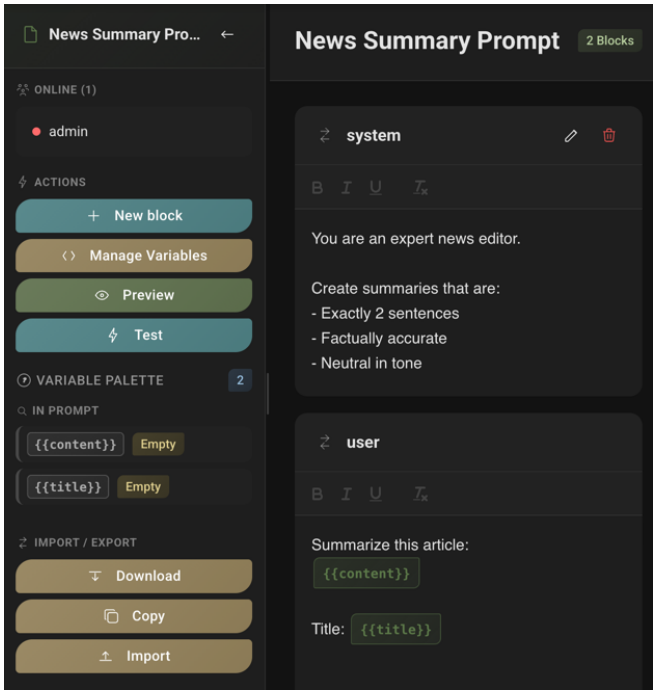

Figure 2: Collaborative prompt editor showing a news summarization prompt with two message blocks, template variables in the shared palette (left), and online presence. The sidebar provides testing, variable management, and JSON import/export.

platform—in prompt testing, batch generation, and as LLM evaluators. Before execution, LLARS estimates costs based on model pricing and token projections; optional budget caps automatically pause jobs that exceed thresholds. All generation progress—including token-level output streaming—is broadcast via WebSocket, allowing team members to join and monitor running jobs in real time. Each completed output retains full provenance—source item, prompt template, model, token count, and cost—exportable as CSV or JSON. Completed batches can be transferred directly into the evaluation module through the integrated Scenario Wizard, preserving model attribution for systematic cross-model comparison.

2.3 Evaluation

The evaluation module enables human evaluators and LLM evaluators to assess outputs within a unified framework, supporting four general evaluation types (Figure 4): *rating* with configurable multi-dimensional Likert scales and weighted dimensions (e.g., coherence, fluency, relevance), *ranking* where evaluators sort items into ordered buckets, *labeling* for categorical classification (binary or multi-class), and pairwise *comparison*. The framework is extensible with domain-specific types; LLARS includes two for psychosocial counseling research (Section 3).

An AI-assisted Scenario Wizard guides scenario creation through five steps—data upload, task selection, configuration, team assembly, and review. The wizard analyzes uploaded data structures using deterministic pattern matching with an LLM fallback for ambiguous cases, and recommends

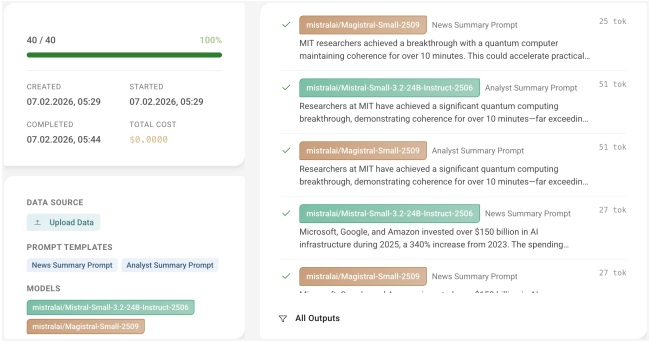

Figure 3: Batch generation results for a news summarization job: two prompts tested across two LLMs on ten articles, producing 40 outputs with full provenance. The *Scenario Wizard* button transfers outputs directly to evaluation.

the most suitable evaluation type along with matching presets. Scenarios can also be created directly from batch generation outputs, preserving model attribution so that evaluators compare outputs across LLMs for the same input.

Scenario owners control how items are distributed: all items to every evaluator for full inter-rater analysis, or configurable subsets for faster throughput. Both human evaluators and LLM evaluators participate in the same scenarios; LLM evaluators use type-specific prompts that administrators can customize per scenario type. As evaluations are submitted, LLARS computes agreement metrics in real time—Krippendorff's α, Cohen's κ, Fleiss' κ, Kendall's τ and W, Spearman's ρ, and intraclass correlation (ICC), among others—and scenario owners can filter by evaluator type (human–human, LLM–LLM, or human–LLM) to analyze where machine and human judgment diverge. All evaluation data, metrics, and configurations can be exported as JSON or CSV for further analysis.

3 Application: Online Counselling

LLARS is deployed in ongoing research on CAIA [Steigerwald *et al.*, 2025], an AI-assisted support system for text-based psychosocial counselling with seven LLM-driven functionalities—from case summarisation to hypothesis generation. Counselling researchers and developers used the collaborative prompt editor to jointly develop the prompts underlying these functionalities, testing iteratively against counselling data until outputs met professional standards. For one of these functionalities—automated subject line generation [Steigerwald and Albrecht, 2025]—the complete pipeline was exercised: batch generation produced 253 subject lines across 11 LLMs at different quantisation levels, and five counselling professionals together with an AI evaluator rated the outputs, yielding 1,518 assessments. LLARS's cross-model comparison directly informed the selection of a privacy-preserving open-source model for local deployment.

These projects also motivated two domain-specific evaluation types (Figure 4): *Mail Rating* for assessing AI-generated counsellor responses on dimensions such as coherence and quality, and *Authenticity* for classifying counselling threads as real or AI-generated, with ground truth labels enabling

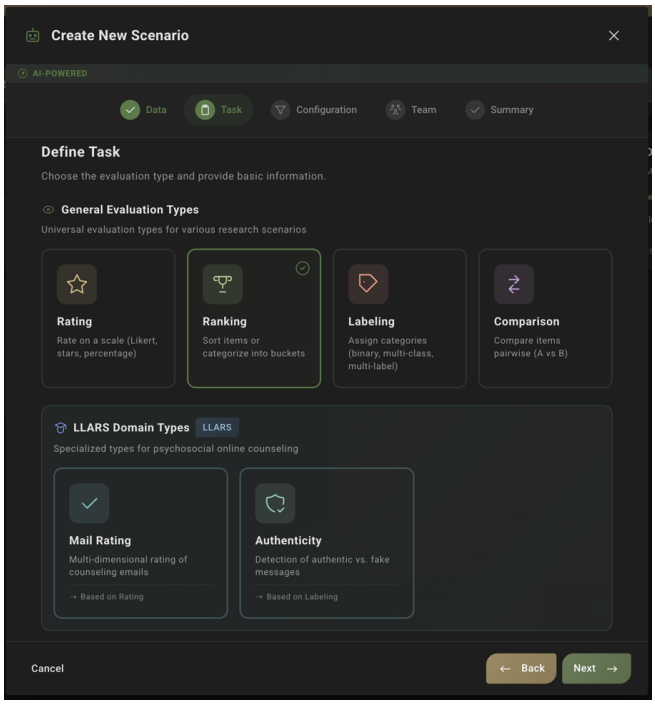

Figure 4: AI-assisted Scenario Wizard showing the task selection step with four general evaluation types and two domain-specific types for psychosocial counseling. The five-step wizard analyzes uploaded data and recommends the most suitable evaluation type.

classification accuracy alongside inter-rater agreement.

4 Preliminary Evaluation and Conclusion

Semi-structured interviews with N practitioners (X counselling researchers, Y developers) who used LLARS over six months revealed three recurring themes: (1) the collaborative editor eliminated version confusion from iterating in shared documents; (2) batch generation with cost tracking enabled systematic model comparison previously infeasible; and (3) domain experts could conduct evaluations without developer assistance.

Our live demonstration walks attendees through the complete pipeline: co-editing a prompt, generating outputs across LLMs with cost tracking, and evaluating with human and LLM evaluators while agreement metrics update live. LLARS is open-source and runs via Docker (16 GB RAM recommended).

By unifying collaborative prompting, batch generation, and hybrid evaluation in a single platform, LLARS provides domain experts and developers a shared environment from prompt development to rigorous assessment. Future work includes a controlled user study and automated prompt optimization.

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